

PAPER_358 — AT2024tvd Wandering Massive Black Hole TDE: Off-Nuclear Disruption Physics

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 97

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Classification: FIRST UQFF off-nuclear wandering massive black hole TDE — frictional timescale and tidal radius

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Abstract

AT2024tvd is the most compelling observed wandering massive black hole (wMBH) caught in the act of tidally disrupting a star at projected physical offset $r_{\text{offset}} = 2.47 \times 10^{17}$ m from the host galaxy nucleus. UQFF computes the tidal radius $r_{\text{tide}} = R_{\text{star}} \cdot (M_{\text{BH}}/M_{\text{star}})^{1/3}$, the dynamical friction timescale t_{fric} for the wMBH sinking to the nucleus, and the full $F_{\text{U_Bi_i}}$ at the off-nuclear disruption site.

2. Core Physics

2.1 Off-Nuclear Disruption Offset

$$r_{\text{offset}} = 2.47 \times 10^{17} \text{ m} \approx 8.0 \text{ pc}$$

This is the projected distance between AT2024tvd and the host nucleus, constraining the wandering distance of the massive black hole.

2.2 Tidal Disruption Radius

$$r_{\text{tide}} = R_{\star} \left(\frac{M_{\text{BH}}}{M_{\star}} \right)^{1/3}$$

For a solar-like star disrupted by a black hole of mass $M_{\text{BH}} \sim 10^6 M_{\odot}$:

$$r_{\text{tide}} = 7 \times 10^8 \times \left(\frac{10^6 M_{\odot}}{M_{\odot}} \right)^{1/3} \text{ m} = 7 \times 10^8 \times 100 = 7 \times 10^{10} \text{ m} \approx 0.5 R_{\odot}$$

2.3 Dynamical Friction Timescale

$$t_{\text{fric}} = \frac{0.428}{\ln \Lambda} \cdot \frac{M_{\text{host}}}{M_{\text{BH}}} \cdot \frac{r_{\text{offset}}^2}{\sigma_{\star}^2} \cdot \frac{1}{r_{\text{offset}}}$$

Simplified:

$$t_{\text{fric}} = \frac{0.428}{\ln \Lambda} \cdot \frac{r_{\text{offset}}}{v_c} \cdot \frac{M_{\text{host}}}{M_{\text{BH}}}$$

For $r_{\text{offset}} = 8 \text{ pc}$, $v_c \sim 200 \text{ km/s}$, $M_{\text{host}}/M_{\text{BH}} \sim 10^3$:

$$t_{\text{fric}} \sim 10^8 - 10^9 \text{ yr}$$

2.4 UQFF $F_{U_Bi_i}$ at Off-Nuclear Site

$$F_{U_Bi_i}(r_{\text{offset}}) = F_{U_Bi_i}(M_{\text{BH}}) \cdot \left(\frac{r_{\text{tide}}}{r_{\text{offset}}} \right)^2$$

The r^2 dependence reflects force dilution with the much larger offset distance.

3. Key Values

Quantity	Formula	Value
r_{offset}	JWST observation	$2.47 \times 10^{17} \text{ m} \approx 8 \text{ pc}$
r_{tide}	$R_{\square} \cdot (M_{\text{BH}}/M_{\square})^{(1/3)}$	$\sim 0.5 \text{ AU}$
t_{fric}	Chandrasekhar formula	$10^8\text{-}10^9 \text{ yr}$
M_{BH}	Spectral fit	$\sim 10^6 M_{\square}$

4. Physical Significance

AT2024tvd is the first confirmed off-nuclear TDE with a massive BH at $> 1 \text{ pc}$ offset. The UQFF framework predicts that the vacuum buoyancy field $F_{U_Bi_i}$ is local — it is set by M_{BH} and r_{tide} , not by the nuclear distance. This means off-nuclear wMBH TDEs have the same $F_{U_Bi_i}$ value as nuclear TDEs of the same BH mass, a testable prediction: UQFF force amplitude should correlate with M_{BH} (not with r_{offset}).

The $t_{\text{fric}} \sim 10^8\text{-}10^9 \text{ yr}$ frictional timescale implies wMBHs are common during the galaxy assembly epoch, and UQFF predicts their spatial distribution modifies the ICM density on 10-100 pc scales.

5. Deduplication Note

- **vs. PAPER_351 (ASASSN-14li):** Both are TDEs; ASASSN-14li is nuclear. AT2024tvd is off-nuclear, introducing r_{offset} and t_{fric} which are absent in PAPER_351.
- **Unique:** First off-nuclear wMBH TDE in the UQFF dataset.

6. Classification

Physics Territory: FIRST UQFF off-nuclear wandering MBH TDE — $r_{\text{tide}} + t_{\text{fric}} + F_{\text{U_Bi_i}}(\text{offset})$

Scale: Sub-galactic (8 pc offset from nucleus)

CP Implementation: AT2024tvdWanderingMBHTDECalculator (CondensedPhysics4.py, Session 97)